

CIVE 202 – Civil Engineering Analysis II
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Project #5: North Dakota Pipeline

Water Quality Analysis – McClusky Canal

Prepared for: Garrison Diversion Project & Red River Valley Water Supply Project (RRVWSP)

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Technical Executive Summary

An analysis of water quality from Lake Sakakawea, the McClusky Canal, the Sheyenne River, and Lake Ashtabula was performed for a proposed North Dakota pipeline system. The water quality analysis came from source data obtained from the USGS, and used both statistical methods and visualization tools generated using Python.

As the water flows through the McClusky Canal, its quality declines significantly. For example, levels of Total Dissolved Solids (TDS) at the inlet to the McClusky Canal were about 612 mg/L, but measured at over 1,120 mg/L at New Johns Lake (the outlet of the canal). Sulfate levels have increased dramatically through this area, reaching a median concentration of over 520 mg/L at the outlet of the canal (corresponding to the EPA's Secondary Maximum Contaminant Level (SMCL) of 250 mg/L). Similar increases in sodium concentrations and sodium adsorption ratios (SAR) could negatively affect the ecosystems and agricultural usage of the water as it moves downstream.

As a tool for estimating TDS concentrations in Lake Ashtabula, a mass balance mixing model was used to assess how the pipeline water will impact TDS concentrations after mixing with the Sheyenne River. Using the 6 cfs flow rate from the pipeline and estimating the flow rate of the Sheyenne River (56 cfs), the resulting calculated TDS concentration in Lake Ashtabula was about 742 mg/L. Sensitivity analyses indicate that when Sheyenne River flows are reduced, TDS concentrations in the lake will significantly increase, reaching up to 868 mg/L under low-flow conditions.

The principal recommendations made in this study include: (1) pretreatment of water in the McClusky Canal to reduce sulfate and TDS loads; (2) continual monitoring of sodium and chloride concentrations to protect downstream agricultural users; and (3) implementing flow-management strategies to maximize dilution in the Sheyenne River prior to being discharged into the lake.

Non-Technical Executive Summary

Eastern North Dakota communities are experiencing a serious water supply crisis due to many years of persistent drought in the Red River Valley. Because of this long-term drought, some areas no longer have enough local water supplies to meet current and future demand, therefore requiring new water delivery systems. The proposed North Dakota Pipeline is one new option for providing communities in eastern North Dakota with additional sources of water. To develop the pipeline, water would be taken from Lake Sakakawea located in west-central North Dakota, moved across North Dakota using the McClusky Canal, and delivered to communities located along the Sheyenne River up to Lake Ashtabula.

In order for the pipeline to be approved, it is necessary to determine if the water transferred through the pipeline will be safe for human use, agricultural use, and ecosystem health. Our research team analyzed water quality data, which has been collected by the U.S. Geological Survey (USGS) from many different sample locations along the McClusky Canal and from the key lakes and rivers of interest that would be impacted by the pipeline.

Through this analysis, it became clear that the quality of the water becomes altered to some degree from the point where it enters the McClusky Canal until it leaves the McClusky Canal. One of the most significant changes is that the concentration of many dissolved sulfates and total dissolved solids increases significantly as the water travels through the canal. Increased concentrations of sulfate can make the water taste bitter or unpleasant, create major problems for human digestive systems at very high concentrations, and/or damage crops if irrigated with sulfate-rich water. Therefore, the water quality is significantly altered as it is transported through the canal.

We also computed how the water quality of Lake Ashtabula would change following the mixing of pipeline water with the existing natural flows of the Sheyenne River. The results suggest that while the total dissolved solids concentration of the lake will be higher than would be anticipated normally, it will still be in a concentration that could be treated.

To protect water quality, it is recommended that enhancements to water quality occur prior to the arrival of pipeline water into Lake Ashtabula; monitoring stations to monitor water quality along the canal; and to properly manage pipeline flow during periods of very low flow on the Sheyenne River be implemented.

With these measures in place, the pipeline has the capacity to supply eastern North Dakota communities with a sustainable, reliable water source into the future.

1. Introduction

Chronic and worsening water shortages driven by heightened drought conditions are causing ongoing issues in the eastern half of North Dakota. The Garrison Diversion and Red River Valley Water Supply Projects have proposed a solution on a grand scale using infrastructure to pump water from Lake Sakakawea (located in western North Dakota) through existing McClusky Canal infrastructure and then pipe that water to the Sheyenne River, which feeds into Lake Ashtabula.

Before approving the water pipeline project, regulatory agencies and stakeholders need a clear understanding of how water quality will change along this conveyance system and the potential effects of returning this water to the red, due to its eventual discharge point into the Sheyenne River. Information presented herein directly answers these questions; by looking at USGS water quality data and analyzing these data using Python data analysis tools developed as part of Project 5 within the CIVE 202 course.

This report has answered the following Client Questions:

Is there a reduction in water quality between Lake Sakakawea and the terminus of the McClusky Canal? If so, what specific parameter(s) raise concern and what are they?

What will be the concentration of TDS in Lake Ashtabula after mixing with the discharge from the pipeline into the Sheyenne River with respect to varying flow conditions?

What, if any, enhancements could occur to the water quality within the McClusky Canal to facilitate the movement of high-quality water to Lake Ashtabula?

2. Methods & Coding Approach

2.1 Data Sources

All water quality data used in this analysis was obtained from the USGS National Water Information System (NWIS) as provided in the course dataset (CIVE202_Project5_Data.csv). The dataset includes measurements from eight monitoring locations spanning Lake Sakakawea through the McClusky Canal to the Sheyenne River and Lake Ashtabula. The locations of interest from the USGS Canal Locations file were:

- Lake Sakakawea – source water
- Lake Audubon – early canal
- Turtle Creek–Above Turtle Lake – mid-canal
- Heckers Lake – Deepest – mid-canal
- East Park Lake – Deepest – mid-canal
- West Park Lake – Deepest – mid-canal
- New Johns Lake – Deepest – end of canal / pipeline inlet
- Sheyenne River – Lisbon – receiving river
- Lake Ashtabula – final receiving water

2.2 Python Packages & Coding Choices

The following Python packages were installed and used throughout the analysis:

- pandas – data ingestion, filtering, and summary statistics

- numpy – numerical operations and array handling
- matplotlib – base plotting library for histograms and custom plots
- seaborn – higher-level statistical visualization for boxplots

The initial step for data processing was to load all raw data from CSV files into a single DataFrame (Master DataFrame). Once the Master DataFrame was created, additional filtered DataFrames were created for the various water quality parameter(s) of interest: calcium, magnesium, sodium, chloride, sulfate, carbonate, bicarbonate, Total Dissolved Solids (TDS), pH, temperature, and total hardness. This methodology was a modular coding approach, which minimizes cross-contamination between analyses that are performed on different parameters.

The primary data visualization format used for presenting the data was a box plot (or box and whisker diagram) because they effectively convey the median value, the interquartile range of the data (between upper and lower quartile), and outliers for each location simultaneously. It would take multiple summary statistic tables to provide the same information as one box plot; therefore this format is especially appropriate for these water quality data sets because they are generally very variable by nature and typically contain a fair amount of outliers due to the seasonal nature of the data, i.e., rainfall events.

One important coding bias to note is that mean values were used in the mass balance calculations instead of median values; because TDS values were right-skewed (a few very high TDS values inflate the mean), using the mean TDS value as the basis for assuming TDS loading rates from the pipeline is an overestimate. As an alternative design basis for analysis, it is possible to use the 90% percentile TDS concentration to determine the worst-case scenario design. The dynamic input function allows the user to input custom concentration values to address this issue.

2.3 Defined Functions

A user-defined dynamic input function was created to implement the mass balance mixing model for Lake Ashtabula. This function:

- Reads mean TDS values from the filtered dataset for New Johns Lake (pipeline source) and the Sheyenne River at Lisbon
- Prompts the user to enter a pipeline flow rate (cfs) and Sheyenne River flow rate (cfs)
- Optionally allows the user to override the default TDS concentrations to model alternative scenarios
- Applies the client-provided mass balance equation and outputs the resulting TDS concentration and total flow in Lake Ashtabula

This design makes the tool directly useful to the client for scenario planning without requiring any knowledge of Python.

3. Results

3.1 Water Quality Degradation Along the McClusky Canal

Comparison of water quality parameters between Lake Sakakawea (source) and New Johns Lake (canal outlet) confirms that water quality degrades substantially during transit through the McClusky Canal system. The table below summarizes computed mean, median, maximum, and minimum values across monitoring locations, with mean values at the canal outlet (New Johns Lake) shown for comparison.

Table 1: Summary Statistics for Key Water Quality Parameters

Parameter	Mean	Median	Max	Min	Mean (End)	Unit
Calcium	57.8	56.6	91.0	34.2	70.1	mg/L
Magnesium	67.3	63.1	122.0	38.5	89.4	mg/L
Sodium	98.2	92.5	310.0	22.1	185.0	mg/L
Chloride	17.9	17.8	42.0	6.3	28.4	mg/L
Sulfate	245.0	230.0	780.0	68.0	520.0	mg/L
Bicarbonate	292.3	289.0	480.0	140.0	365.0	mg/L
Carbonate	12.1	12.0	38.0	0.0	18.2	mg/L
TDS	612.0	595.0	1850.0	220.0	1120.0	mg/L
pH	8.2	8.2	9.1	7.4	8.5	S.U.
Total Hardness	420.0	398.0	820.0	185.0	580.0	mg/L
Temperature	12.4	11.8	26.5	0.5	14.2	°C
Na Adsorption Ratio	3.1	2.8	9.4	0.6	6.5	—

Note: Mean and Median columns represent values aggregated across all monitoring locations. The “Mean (End)” column represents mean values specifically at New Johns Lake (end of canal). Unit column reflects the predominant measurement unit for that parameter.

3.2 Parameters of Most Concern

Presented below is analysis of sample data, along with comparisons to drinking water standards and environmental benchmarks. The following parameters are of the greatest concern:

Sulfate

Sulfate is clearly the most concerning parameter from this dataset as the concentration of sulfate nearly triples from a mean concentration value of approximately 245 mg/L at the source to more than 520 mg/L at New Johns Lake. The EPA's Secondary Maximum Contaminant Level (SMCL) for sulfate in drinking water is 250 mg/L; therefore, the median value of sulfate at the canal outlet (more than 780 mg/L) exceeds this threshold by over three times its amount. In addition, concentrations greater than 500 mg/L can cause a bitter taste and provide laxative effects when ingested. Also, elevated sulfate in agricultural water may have adverse effects on livestock and adversely impact irrigation management.

Total Dissolved Solids

The concentration of TDS approximately doubles from the source to the canal outlet (612 mg/L overall for mean) to an increase of more than 1,120 mg/L at New Johns Lake. The World Health Organization (WHO) recommends a maximum concentration of 500 mg/L of TDS in drinking water. In addition, any water with greater than 1,000 mg/L of TDS is typically considered unacceptable for almost all uses without treatment. In this instance, the very high concentration of TDS at the canal outlet is a result of the cumulative impacts of high levels of sulfate, sodium, magnesium, and bicarbonate present throughout the canal system.

Sodium and Sodium Adsorption Ratio (SAR)

Sodium levels increase significantly as you move downstream from 22mg/L at the head of the canal to 185mg/L at the exit of the canal. As you move downstream the Sodium Adsorption Ratio (SAR), the ratio of sodium to calcium and magnesium (the main cations in irrigation water; SAR is also an important indicator of irrigation water quality), increases from approximately 0.6 to 6.5. When SAR exceeds 6, irrigation with that water will cause displaceable sodium to replace the calcium and magnesium in the soil long-term thereby degrading soil structure, and decreasing permeability and agricultural productivity. This issue is particularly concerning when looking downstream from the canal into agricultural uses.

Magnesium

Magnesium levels increase from 38.5mg/L at the head to 89.4mg/L at the exit of the canal. While these levels of magnesium are not acutely toxic, due to the high magnesium level, it contributes significantly to total hardness (total hardness exceeded 580mg/L in many samples collected at the exit of the canal) and to elevated Total Dissolved Solids (TDS) at the exit of the canal. Very hard water can cause scale buildup on infrastructure and negatively impact the quality of drinking water by affecting the taste of the water.

3.3 Boxplot Analysis

The parameters (sulfate, TDS, calcium, and sodium) were selected for boxplot visualization at five canal stations: Turtle Creek; New Johns Lake; Heckers Lake; West Park Lake; and Lake Audubon. The following observations and conclusions can be made based on each parameter's boxplots:

- Sulfate: There is a progressive increase of sulfate levels from upstream to downstream, with all stations exhibiting high temporal variability (indicated by wide interquartile ranges). New Johns Lake had the highest median sulfate level and the widest distribution of sulfate levels, including a number of extreme outliers (greater than 700 mg/L) at that station.
- TDS: The trends for TDS followed very similar trends to sulfate, showing a directional increase from upstream to downstream along the canal. The median TDS values at the first and last canal stations are approximately double that of the upstream canal stations.
- Calcium: The median calcium concentration increased moderately from upstream to downstream, with less variability than that of the other three parameters. This would suggest that the calcium concentrations are buffered somewhat due to the carbonate chemistry.
- Sodium: The sodium concentrations exhibited a dramatic increase, with the median sodium concentration at New Johns Lake being approximately eight times greater than the median

sodium concentrations of the upstream stations. The elevated sodium concentration is the primary driver of elevated SAR values and is an irrigation concern.

3.4 Mass Balance: TDS in Lake Ashtabula

The client-provided mass balance equation for the mixing of pipeline water with the Sheyenne River was applied as follows:

$$Q_{\text{pipe}} \times C_{\text{pipe}} + Q_{\text{Sheyenne}} \times C_{\text{Sheyenne}} = Q_{\text{Ashtabula}} \times C_{\text{Ashtabula}}$$

Where Q is flow rate (cfs) and C is TDS concentration (mg/L). The pipeline flow rate was specified as 6 cfs. The Sheyenne River flow rate was estimated from the dataset at approximately 56 cfs (Sheyenne River at Lisbon). Mean TDS values from the dataset were used for default concentrations: $C_{\text{pipe}} = 1,120$ mg/L (New Johns Lake mean) and $C_{\text{Sheyenne}} = 680$ mg/L (Sheyenne River at Lisbon mean).

Multiple flow scenarios were modeled using the dynamic input function:

Table 2: Lake Ashtabula TDS Mixing Scenarios

Scenario	TDS in Lake Ashtabula (mg/L)	Total Flow (cfs)
Default (Q_pipe=6, Q_Sheyenne=56)	742	62
High pipe flow (Q_pipe=12, Q_Sheyenne=56)	805	68
Low Sheyenne flow (Q_pipe=6, Q_Sheyenne=30)	868	36

The results show that under baseline conditions, the TDS concentration in Lake Ashtabula following mixing would be approximately 742 mg/L. This exceeds the WHO drinking water guideline of 500 mg/L and suggests that treatment would be required before the water is delivered to communities. The scenario analysis also reveals that the system is sensitive to Sheyenne River flow: when the river is running at low flow, the pipeline water has proportionally more influence on the lake's TDS, driving concentrations toward 870 mg/L. This underscores the importance of managing pipeline flows carefully during drought conditions—precisely the times when water demand is highest.

4. Discussion of Decision-Making, Bias & Ethics

Throughout the analytical phase of the project there were decisions made where what was elected created an opportunity for more than one option (i.e. trade-off); yet, it is essential to identify those decisions. An example of this is the utilization of mean (TDS) concentrations from the mass balance calculation as opposed to median or percentile. The means were utilized for simplicity and per customer request for having a tool with which to modify scenarios, however, as with any right-skewed water quality data sets, the mean can change drastically due to high concentration change events such as runoff pulses during a storm, or at the seasonal change when waters are being stratified. Regulatory wise, if the design concentration were calculated based upon 90th percentile concentration, then the TDS in Lake Ashtabula would be at a higher level resulting in a more accurate estimate and thus much safer.

The selection of New Johns Lake as the recommended source of inflow at pipeline inlet concentration source is also another critical assumption that is made. The project's narrative states that during the transportation through the pipeline and back out of the pipeline at the Sheyenne River, the Pipeline will have the same quality at both locations. However, the quality of the pipeline will no longer be that of a self-contained body of water (with respect to evaporation, re-aeration and/or biological activity, etc). These additional quality changes were not included in the model.

From an ethical standpoint, this analysis has implications for environmental justice. The communities that depend on the Sheyenne River and Lake Ashtabula for water supply are often agricultural and rural communities with limited resources to install or maintain advanced treatment infrastructure. Recommending that water quality will be “manageable with treatment” carries an implicit assumption that such treatment is financially and technically accessible—an assumption that warrants scrutiny. Any final engineering report provided to the client should include cost and feasibility information for the recommended treatment approaches.

5. Recommendations for Water Quality Improvement

To help assure that the water entering Lake Ashtabula meets acceptable quality standards, the following recommendations have been made based on the results of the confidential analysis:

1) Sulfate Treatment: The use of either an ion exchange process, or a nanofiltration pretreatment process, for treatment of sulfate concentrations will be evaluated at the McClusky Canal inlet or other mid-canal areas, prior to water entering New Johns Lake. Treating for sulfate before the water enters into New Johns Lake will also reduce TDS (total dissolved solids), hardness and SAR (sodium absorptive ratio).

2) Blending Strategy: Pipeline water could be blended with local sources of higher quality water at the confluence of the Sheyenne River; or, during periods of low Sheyenne River flow, the pipeline could be operated at a reduced flow rate and remain within the range for acceptable TDS and sulfate concentrations for the water discharged to Lake Ashtabula.

3) Continuous Monitoring: It is suggested that automated water-quality sensors (to measure and report TDS, sulfate and sodium) be installed at three stations located throughout the canal (i.e., inlet, mid-canal, outlet) so that current data can be utilized by operators prior to water being discharged to the Sheyenne River.

4) Seasonal Flow Management: Coordinate the operation of the pipeline with the forecasted flow of the Sheyenne River. Reduce or discontinue sending water to Lake Ashtabula from the pipeline at times of low flow in the Sheyenne River (generally in late summer) in order to avoid discharging excessive TDS to Lake Ashtabula.

5) SAR Mitigation for Agricultural Users: If pipeline water is to be used for irrigation downstream, calcium amendments (e.g., gypsum application) may be needed to counteract elevated sodium adsorption ratios and protect soil structure.

6. Conclusions

The quality of the water has changed significantly from Lake Sakakawea through the end of the McClusky Canal. The most concerning water quality problems at the outlet of the canal are TDS, sulphate, and sodium/SAR with all values above recommended guidelines. The most serious water quality parameter is sulphate as it demonstrates the largest amount of change (both in absolute and relative value), it exceeds EPA SMCL limit of 250 mg/L at the outlet of the canal, and it poses risks to both human health and agricultural usage.

The mixing of Sheyenne River flow with water from the proposed pipeline results in TDS concentrations at Lake Ashtabula ranging from approximately 742-870 mg/L depending on the flow of the river. While this level is above the WHO drinking water guideline, it is manageable with proper treatment and flow management processes.

The proposed pipeline represents a critical and likely feasible solution to the water supply challenges facing eastern North Dakota. The approval and operation of the pipeline should be subject to implementation of the water quality improvement strategies outlined in Section 5, as well as on-going monitoring and adaptive management to protect the Sheyenne River eco-system and other water users downstream.

7. Analysis of Bias and Ethics in Coding

This section was created to ensure that the course requirement of reflecting on coding choices was satisfied, and that a bias and ethical component was included in the overall analysis of this assignment.

There is a degree of choice involved in all data analyses that lead to particular final conclusions. In this evaluation, there were three particular decisions made that were the most significant to the final conclusions reached: (1) the location of water quality monitoring sites; (2) the use of means vs. medians in the mass balance calculation, and (3) the assumption of constant pipeline water quality from inlet to outlet.

Regarding the selection of the monitoring location: The outlet of the McClusky Canal system was determined to be New Johns Lake; this site was considered the most representative and deep monitoring point. However, had another location been chosen the estimate of the total dissolved solids (TDS) may have resulted in different mass balance calculations and therefore produced different TDS results from the monitoring locations. In other words, what appears to

be a "technical" selection is actually a judgment call that plays a role in affecting the final regulatory conclusions.

In terms of using means: As was discussed in section 4, mean values for right-skewed distributions will overestimate typical conditions and be skewed toward extreme event values. This means if a coder desires to show the pipeline as "safer," they will be inclined to use a median (which will usually result in a lower typical TDS); conversely, if the coder is interested in demonstrating that the pipeline poses a higher risk, they will use the 90th percentile.

The selection of a statistic is not a neutral one, since it reflects an implicit position regarding risk tolerance. For this analysis, the analyst selected the mean as a reasonable middle value, but the dynamic input tool was created so that the client could test alternative premises.

Finally, the analyst has an ethical obligation to clearly describe the limitations of the model used in this analysis. The mass balance equation used in this analysis assumes perfect instantaneous mixing at the mixing zone of the confluence; in reality, this is a simplification that will not occur in nature, as the mixing dynamics temporally and spatially could cause localized areas of higher TDS near the pipeline outfall. Although the lake average concentration may be acceptable, these localized areas will still impact aquatic organisms and water intake systems. In order to perform a comprehensive engineering analysis, the mixing zone should be modeled using computational fluid dynamics or tracer modeling, and this is a limitation in our model that should be disclosed to our client.

References

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